

K-Means clustering

K-Means clustering is an iterative, unsupervised machine learning algorithm designed to partition unlabeled data into **K distinct, non-overlapping groups (clusters)** based on feature similarity. The core objective of the algorithm is to minimize the variance within each cluster, effectively grouping data points so that items in the same cluster are as similar as possible while remaining distinct from other groups.

Core Concepts

- **The "K":** This represents the total number of predefined groups or clusters you want to create. You must specify this number before running the algorithm.
- **Centroids:** These are the imaginary or real data points representing the "centre" of each cluster. Every cluster is anchored by exactly one centroid, which is recalculated as the mean of all points assigned to that cluster.
- **Distance Metric:** To measure similarity, K-Means typically calculates the **Euclidean distance** between a data point and a centroid.

The Step-by-Step Algorithm

The algorithm relies on an optimization process known as Expectation-Maximization

1. **Choose the number of clusters (K):** Define how many groups you want to find.
2. **Initialize Centroids:** Randomly pick K data points from the dataset to serve as the initial cluster centres (centroids). Alternatively, modern implementations use advanced setups like [greedy k-means++](#) to spread out initial centroids and speed up processing.
3. **Assign Points (Expectation Step):** Calculate the distance from every data point to all K centroids. Assign each data point to its nearest centroid's cluster.
4. **Update Centroids (Maximization Step):** Recalculate the position of each of the K centroids by taking the average (mean) of all coordinates of the data points assigned to that specific cluster.
5. **Repeat until Convergence:** Run steps 3 and 4 repeatedly. The algorithm stops when [centroids stop changing positions](#), data points stop switching clusters, or a maximum iteration limit is reached.

Mathematical Objective

K-Means works to minimize a cost function known as the **Within-Cluster Sum of Squares (WCSS)** or inertia.

$$WCSS = \sum_{j=1}^K \sum_{i \in S_j} ||x_i - \mu_j||^2$$

- K is the number of clusters.
- S_j is the set of points belonging to cluster j.
- x_i is a specific data point.
- μ_j is the centroid of cluster j

Methods to Calculate the Optimal K

Because K-Means requires the number of clusters K to be specified in advance, data scientists use several mathematical heuristics to determine the ideal value.

1. The Elbow Method

- Runs K-Means across a range of sequential K values (e.g., 1 to 10).
- Plots the **Within-Cluster Sum of Squares (WCSS)**, also called **inertia**, against K.
- Identifies the "elbow" point where the rate of decline flattens out sharply.
- Selects this inflection point as the optimal K value.
- Learn more about this graphical tool on the [GeeksforGeeks Elbow Method Guide](#).

2. Silhouette Analysis

- Measures how close each point in one cluster is to points in the neighbouring clusters.
- Computes a **Silhouette Coefficient** ranging from -1 to +1 for each point.
- Scores near +1 indicate the point is far away from neighbouring clusters (well-clustered).
- Scores near 0 indicate the point is on or very close to the decision boundary.
- Averaging these scores across different K values reveals the configuration with the cleanest separation.

3. Gap Statistic Method

- Compares the total intra-cluster variation for different values of K with their expected values under a null reference distribution of the data.
- Standardizing this measurement isolates the value of K that yields the largest gap relative to random distribution.
- Evaluates the cluster boundaries with high statistical precision, though it can be computationally slow.

4. Davies-Bouldin Index

- Calculates the average similarity measure of each cluster with its most similar cluster.
- Similarity is defined as a ratio of within-cluster distances to between-cluster distances.
- Lower index scores signify better clustering, indicating tight, highly separated categories.

Limitations of K-Means Clustering

While highly efficient and simple to scale, K-Means possesses structural limitations that can degrade performance on real-world datasets.

- **Predefined K Requirement:** The user must explicitly guess or calculate K before running the model, which can lead to artificial or illogical groupings if miscalculated.
- **Sensitivity to Initial Centroids:** Randomly placing initial centroids can trap the algorithm in a **local minimum** rather than a global optimum. This is often mitigated using initialization variants like **K-Means++**.
- **Assumption of Spherical Clusters:** The algorithm assumes clusters are circular/spherical and of similar density and size. It fails drastically when faced with complex, elongated, or non-convex geometries.
- **Extreme Sensitivity to Outliers:** Outliers and anomalies skew the cluster averages because centroids are calculated as a simple arithmetic mean. A single distant point can heavily distort the final boundaries.
- **Curse of Dimensionality:** As the number of features grows, distance metrics like Euclidean distance break down because the distance between any two vectors converges. Dimensionality reduction (such as PCA) is often required beforehand.